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Department of Information and Communication Technology (ICT)

4th Year 1st Semester B.Sc.(Engg.) Final Examination 2024

Course Title: Cryptography and Cyber Law Course Code: **ICT – 4105**

Total Marks: **70** Time: **3 hours**

Answer any five (05) sets of the following questions

1. (a) Briefly describe the following terms: Access Control, Authentication, Data Integrity, Nonrepudiation, and Traffic Analysis. **5**

(b) Explain the relationship between security services and mechanisms. **3**

(c) Describe the active attacks with suitable figure. **4**

(d) What is the difference between an unconditionally secure cipher and a computationally secure cipher? **2**

2. (a) Prove Fermat's Little Theorem and explain how it is used to test for primality. Is 561 a prime number based on this test? Evaluate $5^{123} \ mod\ 175$ using Fermat's Little Theorem. Show all steps. **4**

(b) State and prove the Chinese Remainder Theorem. Then solve the following system of congruences: **4**

$$x = 2\ mod\ 3, x = 3\ mod\ 5, x = 2\ mod\ 7$$

(c) Determine the multiplicative inverse of 7 mod 26 , if it exists using extended Euclidean algorithm. **3**

(d) Find x such that $43x \equiv 1\ mod\ 240$ **3**

3. (a) Explain how legal frameworks such as the General Data Protection Regulation (GDPR) help mitigate cyber-attacks and protect user privacy. **2**

(b) Explain the basic working of the DES algorithm using a simple 64-bit plaintext block and a 56-bit key. Show how the initial permutation, round functions, and final permutation contribute to the encryption process. **6**

(c) In the DES algorithm, a 64-bit plaintext block is divided into two 32-bit halves: L_0 and R_0. Given $R_0 =$ 0xF0F0F0F0 and round key $K_1 =$ 0x0F0F0F0F , compute the output of the first round's function $f(R_0, K_1)$ assuming XOR operation only. Then, find $L_1 = R_0$ and $R_1 = L_0 \oplus f(R_0, K_1)$, where $L_0 =$ 0xAAAAAAAA . **6**

4. (a) Which AES modes cause error propagation during decryption, and how does this affect the integrity of the decrypted message? Illustrate with CBC and CFB modes. **2**

(b) Given the input word, [0x23,0xA7,0x4C, 0x19]. Use the following partial AES S-box to perform the *SubBytes* transformation. Provide the resulting output word. **6**

Row ↓ Col →	3	4	7	9	A	C
1	6D			C6		
2	D4					
4		A1				2E
A			63		D2	

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(c) In AES encryption, apply the AddRoundKey step only. Given: Input word: $[0x1A, 0x2B, 0x3C, 0x4D)$ Round key word: $[0x55, 0x66, 0x77, 0x88)$ Perform XOR between the input word and the round key, and write the resulting output word. 6

5. (a) Explain the concept of side-channel attacks and provide one example of a timing attack on RSA. 4

(b) Consider a Diffie-Hellman scheme with a common prime q = 11 and a primitive root α = 2. 5

I. Show that 2 is a primitive root of 11.
II. If user A has public key YA = 9, what is A's private key XA?
III. If user B has public key YB = 3, what is the shared secret key K, shared with A?

(c) Discuss three hardware and software countermeasures for mitigating side-channel vulnerabilities. 5

6. (a) Explain the layered architecture (protocol stack) of SSH. Briefly describe the roles of each layer 4

(b) A simple (non-cryptographic) hash function $H(x)$ is defined as the sum of ASCII values of the characters in a message, modulo 100: 5

$$H(x) = \left(\sum \text{ASCII of characters in x}\right) mod\ 100$$

Compute the hash value of the message "AB" and "BA" using this function. Do the two messages produce the same hash? What does this imply about collision resistance in weak hash functions?

(c) In a secure message system, a simple Message Authentication Code (MAC) is computed using modular addition: $MAC = (Message + SecretKey) mod\ 17$. Suppose, *Message*: 15, *Secret key*: 7. Compute the *MAC* for the message. Suppose an attacker changes the message to 10 but doesn't know the key. Can they forge the correct MAC easily? Explain briefly. 5

7. (a) How does ECC achieve the same level of security as RSA with a smaller key size? Briefly explain. 2

(b) Given the elliptic curve $y^2 = x^3 + 2x + 3\ mod\ 97$, determine whether the point $P = (3,6)$ lies on the curve. 6

(c) Given public key ($p = 23$, $g = 5$, $h = 8$) and message $m = 10$, compute the ElGamal ciphertext using random $k = 6$. 6

8. (a) What is ICT ethics? Why is it important in the digital age? 4

(b) What are the fundamental ethical concepts that guide decision-making in ICT? 4

(c) What responsibilities do ICT professionals have towards their clients, employers, and the public? 3

(d) What do you mean by ethical hacking? How is it different from malicious hacking? 3

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